

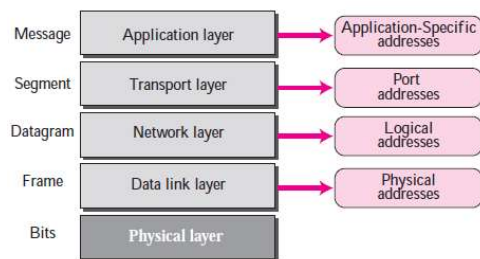
Types

- IP
- TCP
- UDP
- HTTP
- ICMP
- ...
- ...
- ...

Logical Address vs. Physical Address

MAC vs IP address

MAC address	IP address
Functions at data link layer	Functions at network layer
It is a physical address	It is a logical address
It is fixed	It changes with the relocation of device from one network to another
It is a 48 bit address	It is a 32 bit address



Internet Protocol (IP)

Internet Protocol is the primary network protocol used on the Internet, developed in the 1970s. On the Internet and many other networks, IP is often used together with the Transport Control Protocol (TCP) and referred to interchangeably as TCP/IP

IP supports unique addressing for computers on a network. Most networks use the Internet Protocol version 4 (*IPv4*) standard that features IP addresses four bytes (32 bits) in length. The newer Internet Protocol version 6 (*IPv6*) standard features addresses 16 bytes (128 bits) in length.

- IPv4 uses 32 binary bits to create a single unique address on the network. An IPv4 address is expressed by four numbers separated by dots. Each number is the decimal (base-10) representation for an eight-digit binary (base-2) number, also called an octet. For example: 216.27.61.137

IPv4

An IPv4 address (dotted-decimal notation)

172 . 16 . 254 . 1

↓ ↓ ↓ ↓

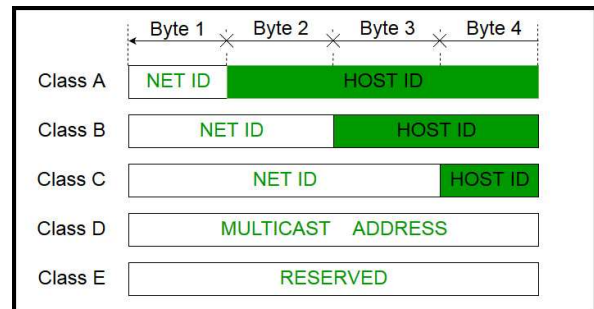
10101100 . 00010000 . 11111110 . 00000001

One byte = Eight bits

Thirty-two bits (4 x 8), or 4 bytes

- IPv6 uses 128 binary bits to create a single unique address on the network. An IPv6 address is expressed by eight groups of hexadecimal (base-16) numbers separated by colons, as in 2001:cdba:0000:0000:0000:0000:3257:9652. Groups of numbers that contain all zeros are often omitted to save space, leaving a colon separator to mark the gap (as in 2001:cdba::3257:9652).

Classes of IP Address



IP Address Range

IP Address Class	IP Address Range (First octet decimal value)
Class A	1-126 (00000001 to 01111110) *
Class B	128-191 (10000000 to 10111111)
Class C	192-223 (11000000 to 11011111)
Class D	224-239 (11100000 to 11101111)
Class E	240-255 (11110000 to 11111111)

Class A Extremely large network, 16,777,000 available host address.
Class B Large network, 16,384 networks, each supporting more than 65,000 hosts.
Class C small networks, more than 2 million networks, each supporting up to 256 host.
*127 (01111111) is a Class A address reserved for loopback testing and cannot be assigned to a network. (127.0.0.1)

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Class	Subnet Mask decimal	No. of Hosts per Network	No. of Networks	Start - End Address
A	255.0.0.0	16 Million	127	1.0.0.0 - 126.255.255.255
B	255.255.0.0	65000	16000	128.0.0.0 - 191.255.255.255
C	255.255.255.0	254	2 Million	192.0.0.0 - 223.255.255.255
D	Reserved for multicast groups			224.0.0.0 - 239.255.255.255
E	Reserved for future use, or Research and Development Purposes			240.0.0.0 - 254.255.255.254

IP Address **10.10.10.10**
Subnet Mask **255.0.0.0**
IP Address **172.168.10.1**
Subnet Mask **255.255.0.0**
IP Address **192.168.1.1**
Subnet Mask **255.255.255.0**
*Network portion *Host portion

IPv6

An IPv6 address (in hexadecimal)

2001:0DB8:AC10:FE01:0000:0000:0000:0000

↓ ↓ ↓ ↓ ↓
2001:0DB8:AC10:FE01:: Zeroes can be omitted

10000000000001:0000110110111000:1010110000010000:1111111000000001:
0000000000000000:0000000000000000:0000000000000000:0000000000000000

Data on an Internet Protocol network is organized into *packets*. Each IP packet includes both a header (that specifies source, destination, and other information about the data) and the message data itself.

IP functions at layer 3 of the OSI model. It can therefore run on top of different data link interfaces including Ethernet and Wi-Fi.

- At the dawn of IPv4 addressing, the Internet was not the large commercial sensation it is today, and most networks were private and closed off from other networks around the world. When the Internet exploded, having only 32 bits to identify a unique Internet address
- Under IPv4, there are 232 possible combinations, which offers just under 4.3 billion unique addresses. IPv6 raised that to a panic-relieving 2128 possible addresses.

IP Header

4 Bits	8 Bits	16 Bits	24 Bits
Version	IHL	Type of Service	Total Length
Identification		Flags	Fragment Offset
Time to Live	Protocol	Header Checksum	
Source IP Address			
Destination IP Address			
IP Options			Padding
Data			

- Version (Specifies the format of the IP packet header): 4 bits.
- IHL, Internet Header Length (Specifies the length of the IP packet header in 32 bit words. The minimum value for a valid header is 5): 4 bits.
- Type of Service (The Type of Service provides an indication of the abstract parameters of the quality of service desired.): 8 bits.

Contd...

- Total length (Contains the length of the datagram). 16 bits.
- Identification (Used to identify the fragments of one datagram from those of another. The originating protocol module of an internet datagram sets the identification field to a value that must be unique for that source-destination pair and protocol for the time the datagram will be active in the internet system.): 16 bits.

Contd...

- Flags: (Various Control Flags.) 3 bits
 - Bit 0: reserved, must be zero
 - Bit 1: (DF) 0 = May Fragment, 1 = Don't Fragment.
 - Bit 2: (MF) 0 = Last Fragment, 1 = More Fragments.
- Fragment Offset (Used to direct the reassembly of a fragmented datagram): 13 bits.
- TTL, Time to Live (A timer field used to track the lifetime of the datagram. When the TTL field is decremented down to zero, the datagram is discarded): 8 bits.

Contd...

- Protocol (This field specifies the next encapsulated protocol): 8 bits.
- Header checksum (A 16 bit one's complement checksum of the IP header and IP options): 16 bits.
- Source IP address (IP address of the sender): 32 bits.
- Destination IP address (IP address of the intended receiver): 32 bits.
- IP Options (variable)
The options may appear or not in datagrams. They must be implemented by all IP modules
- Padding is used to make sure that the IP packet header has a length that is a multiple of 32 bits.

TCP Header Format

0	4	10	16	24	31
SOURCE PORT			DESTINATION PORT		
SEQUENCE NUMBER					
ACKNOWLEDGMENT NUMBER					
HLEN	RESERVED	CODE BITS		WINDOW	
CHECKSUM			URGENT POINTER		
OPTIONS (IF ANY)				PADDING	
DATA					
...					

Transmission Control Protocol (TCP)

Transmission Control Protocol (TCP) and Internet Protocol (IP) are two distinct network protocols, technically speaking. TCP and IP are so commonly used together, however, that TCP/IP has become standard terminology to refer to either or both of the protocols.

- IP corresponds to the Network layer (Layer 3) in the OSI model, whereas TCP corresponds to the Transport layer (Layer 4) in OSI. In other words, the term TCP/IP refers to network communications where the TCP transport is used to deliver data across IP networks.

UDP header format

0	15	16	31
Source port		Destination port	
UDP length		Checksum	

Datagram

- A **datagram** is a basic transfer unit associated with a packet switched network in which the delivery, arrival time, and order of arrival are not guaranteed by the network service.
- Each datagram has two components, a header and a data payload. The header contains all the information sufficient for routing from the originating equipment to the destination without relying on prior exchanges between the equipment and the network. Headers may include source and destination addresses as well as a type field. The payload is the data to be transported. This process of nesting data payloads in a tagged header is called encapsulation.